1. Problem Definition and Project Scope

Organizations often face difficulties understanding how a change in one business process can affect other activities, approvals, and final results. Employees and managers may need to check these connections manually, which takes time and can lead to missed steps or unexpected problems. A software solution can help by showing the relationships between activities and allowing users to test proposed changes before applying them to actual operations. The intended users are process managers, business analysts, and employees responsible for reviewing workflows. The proposed Process Change Impact Simulator will allow users to create or import process models, identify dependencies, analyze the possible effects of changes, and compare original and modified workflows. The system will use Python, Flask, SQLite, and graph algorithms to perform these tasks. The project will focus on sample workflows, dependency analysis, and basic process simulation. It will not modify live organizational systems, make automatic business decisions, or guarantee predictions about every real-world consequence.

2. Users and Stakeholders

The Process Change Impact Simulator will have four main user categories.

1. Process Analyst: The process analyst will create or import business process models containing activities, decision points, approval conditions, and dependencies. The analyst will select an activity or rule for modification and run impact analysis to identify affected downstream activities and process paths. The system should provide a simple interface for defining and reviewing workflows.

2. Process Manager: The process manager will review proposed process changes before they are applied to actual operations. The main goal is to understand how a change may affect approvals, required activities, and final outputs. The manager will compare the original and modified workflows and use the results to support decisions.

3. System Administrator: The administrator will maintain the application, manage stored process data, and check whether the system is functioning correctly. The administrator may also manage sample process files and support database maintenance.

4. Organization or Business Owner: This stakeholder is interested in reducing workflow errors and understanding the possible consequences of process changes. The owner may review simulation reports to assess whether a proposed change could affect business operations.